Transverse cross-norm and a marked inverse-affine action

Cross-difference norm -> short normalized base

N_Q / c_Q = x((1-pi)P)-x((1-pi)R); the base is identical for Q and -Q

Square-zero first variations -> graph-dependent coefficients

One zero factor at R=P+U; retain B and C, without dividing by that factor

Ratio and rational lift -> inverse-affine point action

C/B = x(P+[(1-k)^-1]U); subtract P and choose the rational lift

Reusable first call -> ordinary marked action

bG=M_Q(G), Q2=G-bG, output U-M_Q2(U); charge both constructors

Construction, 2026-11-08, T-1931. This round constructs a short zero-order cross-norm from the Frobenius relation and isolates its target-dependent first variations. Their ratio gives a marked inverse-affine point action, which a costed two-call bridge converts into the ordinary[k]U action. Complete native controls cover422 inverse-affine actions and492 ordinary actions;24 selected P-192 blocks bind the same local coefficient interface to the primary curve. These are written derivations and unsigned producer checks, pending trusted review. [pcc:unverified]

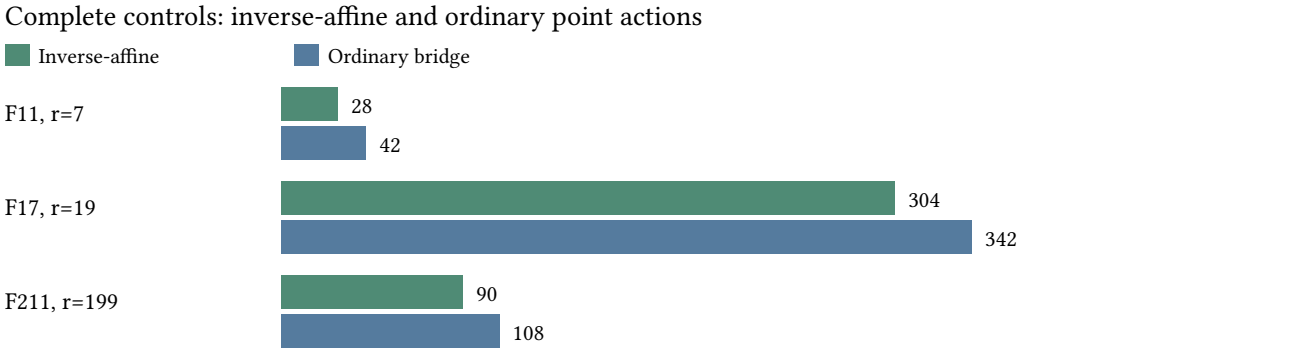
Cross-norm identity. Assume E(F_p)=<G> has odd prime order r!=p, Q=[k]G is nonzero and Q!=+/-G. Put phi=1-pi and N_Q(P,R)=product_j(x(P+jG)-x(R+jQ)). A divisor calculation gives N_Q=c_Q*(x(phi(P))-x(phi(R))), where c_Q=product_(j!=0)(x(jG)-x(jQ)) is nonzero. The zero divisors correspond to P-R in E(F_p) and P+R in E(F_p); both rational translation families are covered once because Q-G and Q+G generate the group. Poles and leading terms determine the identity and constant. cross-norm-proof.md contains the full argument. [pcc:unverified]

Compressed base and the graph mark. The normalized base N_Q/c_Q has the short expression obtained from x(phi(P))=((Y^p+Y)/(X^p-X))^2-X^p-X. The direct c_Q constructor still costs r-1 factors. Its sign and inverse symmetries are proved by reindexing and checked60 times. All90 frozen generic probes match the Frobenius expression. Each declared curve panel has one observed constant:1 over F11,13 over F17 and82 over F211. This observation is not promoted to universal k-independence. In particular N_Q=N_(-Q) as rational functions, so the base function and all its P,R derivatives alone do not retain the sign of the graph. [pcc:unverified]

Transverse coefficients. In K[s,t]/(s^2,st,t^2), write product_j(d_j+s+t*a_j)=A+s*B+t*C, with d_j=x(P+jG)-x(R+jQ), a_j=x(P+jG). At R=P+U for rationalU and2P nonrational, there is exactly one zero factor, indexed by[j](G-Q)=U. The opposite-sign equality would force2P rational. Thus B is nonzero and C/B=x(P+jG). Lift this x-coordinate, subtractP and select the unique rational point to obtain M_Q(U)=[(1-k)^-1]U. This includes U=O. The three-coefficient algebra uses only first variations, with no double-pole Laurent precision or inversion of a zero factor. [pcc:unverified]

Ordinary action bridge. For the nonexceptional public target, compute bG=M_Q(G), set Q2=G-bG, and return U-M_Q2(U). Since b=(1-k)^-1, the second inverse-affine multiplier is1-k, so the result is[k]U. The first action and Q2 setup are reusable across later inputs. Public point tests handle Q=G,-G and[2]Q=G by known multipliers1,-1,(r+1)/2; Q=O is the separate trivial written branch. The executable panels cover the declared nonzero targets. Constructor, first action, second constructor and per-input action remain charged. [pcc:unverified]

Exact complete controls. The F11,r7 and F17,r19 panels use all nonzero targets and all inputs includingO. The F211,r199 panel uses twelve frozen targets and nine inputs. There are422 inverse-affine actions, including30 zero inputs, and392 reversed-second-graph checks with distinct nonzero-input outputs. The bridge gives492 ordinary actions:396 second-stage norm calls,28 reusable setup norm calls and96 public special-case outputs. The streaming square-zero product and a prefix/suffix cofactor calculation reproduce every retained coefficient. Their field and curve helpers are shared; their norm contractions differ. [pcc:unverified]



Source: norms.json. Separately: 24 selected P-192 blocks, not a full P-192 norm.

Public transfer check and costs. A separate checker uses only delta=G-Q and U: hash[b]delta with[b]G and scan U-[im]delta while retaining[im]G, m=ceil(sqrt(r)). It agrees with all1238 norm calls, including negative controls and bridge calls, using4310 giant lookups in that check run. It never reads a scalar label. The generic transfer algorithm uses O(sqrt(r)) point work and memory with a reusable table; the checker rebuilds tables and its extra work is charged. The direct complete graph norm uses O(r) point work and field contraction, with stored graphs/arrays. A streaming contraction can reduce working storage, but the implemented graph construction is explicit. No timing comparison or speedup is claimed. [pcc:unverified]

P-192 local binding. Keep the standard Round52 curve/public target, the two Round57 CM fixtures, and P=6S from Round54. Use the sixteen indices0..15 and eight marked indices0,1,2,3,5,7,11,15. For each target construct U=[j](G-Q) from public points. The24 selected blocks have one zero factor, nonzeroB, and their ratio decodes to[j]G; three controls have U=O. The Frobenius normalizer maps P and P+U to the same point. Both norm methods and fresh-copy replays agree. These are selected blocks, not a full P-192 norm, arbitrary-input inverse action or singular-index locator. [pcc:unverified]

What must be compressed next. The base N_Q=N_(-Q) is identical while the marked outputs differ for every nonzero input in the392 sign checks. Therefore differentiating only the short base expression in its original point variables cannot supply the transverse coefficients. The next evaluator must carry the graph-dependent square-zero deformation. The paired factor formulas in cross-norm-proof.md give an explicit interface retaining the relative sign; this round's native code directly multiplies unpaired factors. The principal CM relation1-pi provides the compressed base. A smaller class number or L-value has not supplied the B,C evaluator here; first-variation compatibility under the costed horizontal relations is the next concrete construction. [pcc:unverified]

Verification and custody. Scope and source commits precede affected checks, and a before-data amendment adds the public transfer checker. Four fresh native outputs agree; a changed retained coefficient is rejected with exit101. Source remains native Rust with locked dependencies. The patch-application mismatch was corrected before building the added control; no result was overwritten. validation.json records the required legacy library check and actual failures. No cloud job or generator was started or modified. The mathematical argument is written rather than a new Lean formalization, and no trusted PCC promotion is asserted. [pcc:unverified]

Sources checked (KB unavailable). The MCP tools are not loaded and the migrated wrapper is absent. Stacks0BIE, https://stacks.math.columbia.edu/tag/0BIE , supplies the norm construction. BFLS, https://arxiv.org/pdf/2003.10118 , Example4.13 and section4.15 supply jet-algebra and addition-compatibility background. Dang-Moody2025, https://doi.org/10.1007/s10623-025-01692-y , equations2-4 and section3 provide prior sum/product/symmetric isogeny formulas; their general computational cost is not imported into this coupled norm. The DOI fetch failed, while the publisher's indexed primary text supplied those excerpts. Standard product formulas are prior art; no novelty claim for those ingredients is made. [pcc:unverified]